

# Binder Remodelling with Boltz

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# 1. Prerequisites

## 1.1 Install VS Code

Download and install VS Code from the official download page:

[https://code.visualstudio.com/download?exp\\_download=fb315fc982](https://code.visualstudio.com/download?exp_download=fb315fc982)

## 1.2 Github Account

Create a GitHub account at <https://github.com/> if you do not already have one.

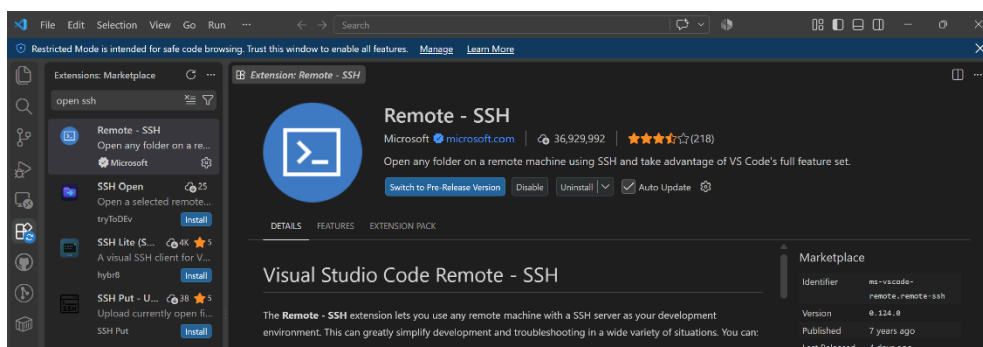
## 1.3 La Trobe VPN

To securely connect to Ewok machine off campus, download and setup the VPN here:

<https://www.latrobe.edu.au/students/support/it/connecting/vpn>

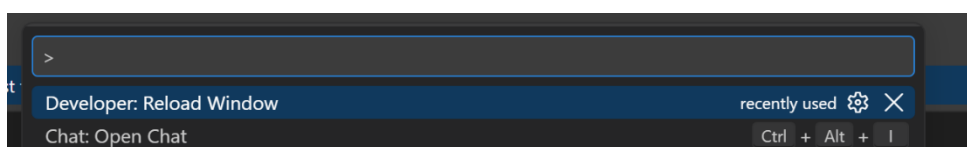
## 1.4 Remote- SSH extension

Install the **Remote – SSH** extension in VS Code. This extension enables you to connect to an Ewok machine from your local computer and access its computational resources directly through VS Code.



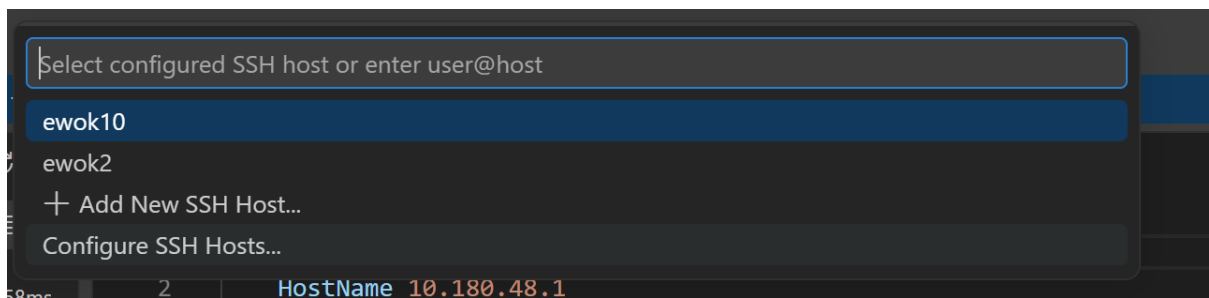
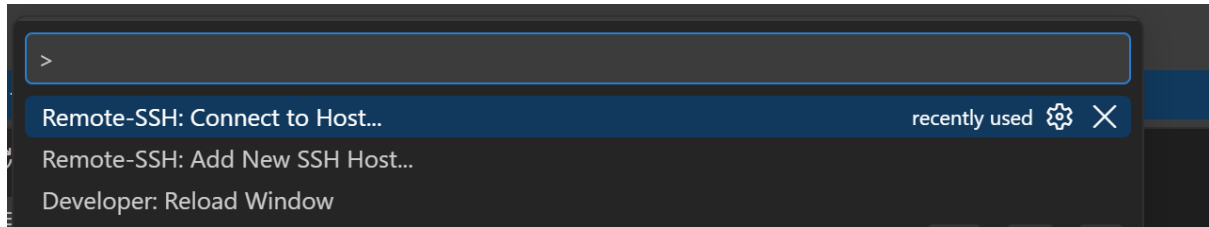
After installation, open the **Command Palette** (**Ctrl + Shift + P** on Windows or **Cmd + Shift + P** on Mac.) and type **>Developer: Reload Window**, and press Enter to ensure the extension loads correctly.

**Quick tip:** To run any command in the **Command Palette**, make sure to start it with a **>** symbol, e.g. **>your command**.



## 2. Connect to the Ewok Machine

In VS Code, open the **Command Palette** and type **>Remote-SSH: Connect to Host**, then select **Configure SSH Hosts**.



Add an entry for your Ewok machine using the following configuration format:

Host <ewok\_machine\_number>

HostName <your\_ewok\_machine\_ip\_address>

User <your\_ewok\_username>

Port <your\_ewok\_port>

Example configuration:

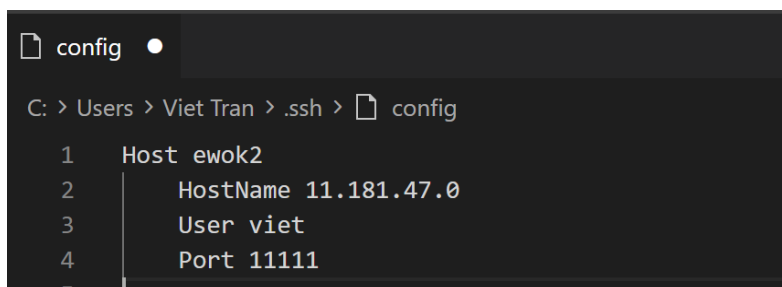
Host ewok2

HostName 11.181.47.0

User viet

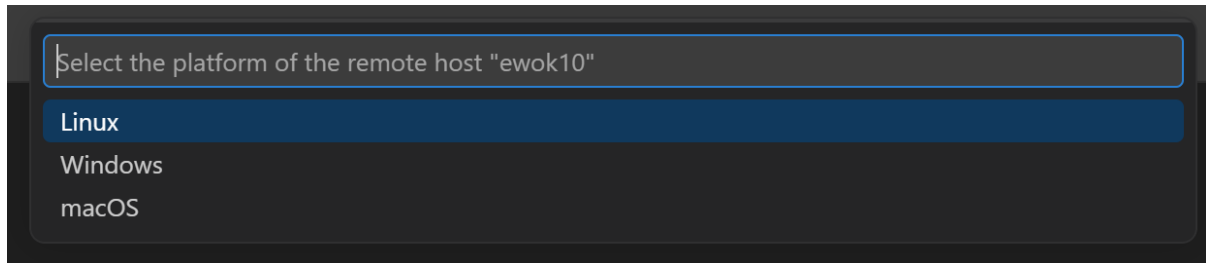
Port 11111

Make sure to save your changes to the config via **Ctrl + S** on Windows or **Cmd + S** on Mac.

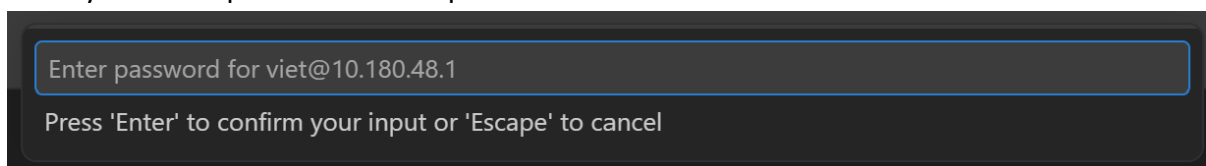


**Note:** Contact [Zhen He](#) or [Aaron Harris](#) for setting up and access your Ewok account.

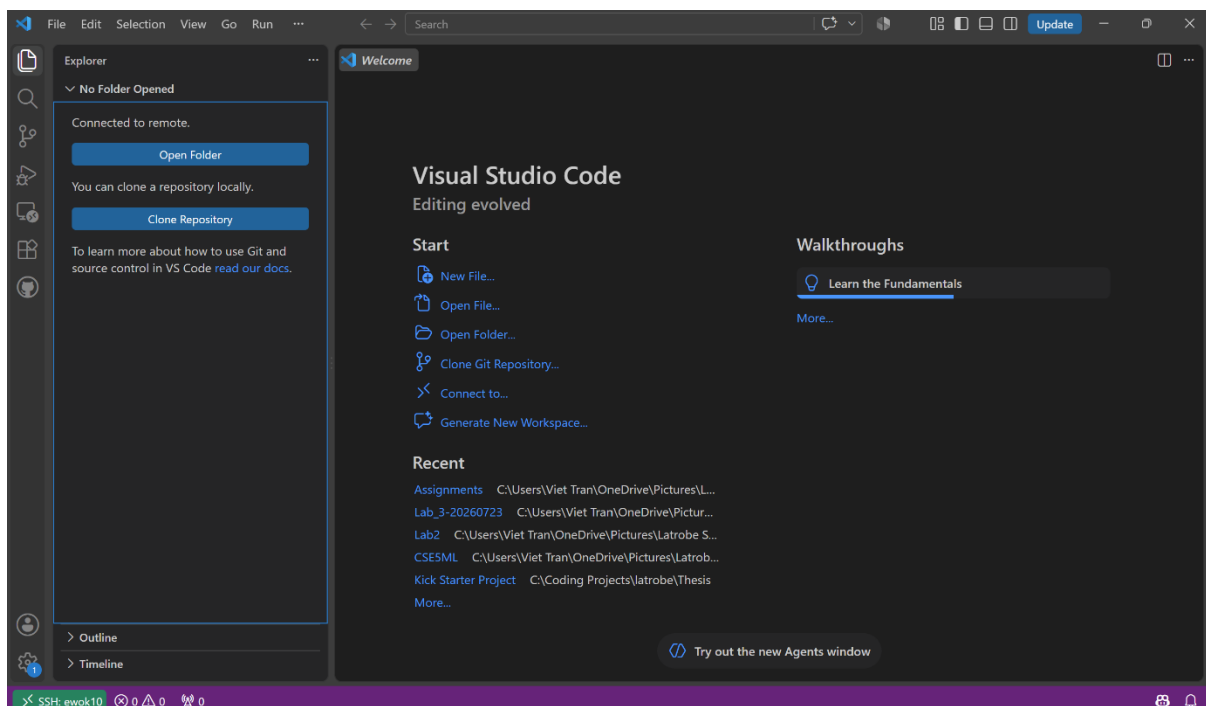
If VS Code prompts you to select the remote host's operating system, choose **Linux**



Enter your Ewok password to complete the connection.



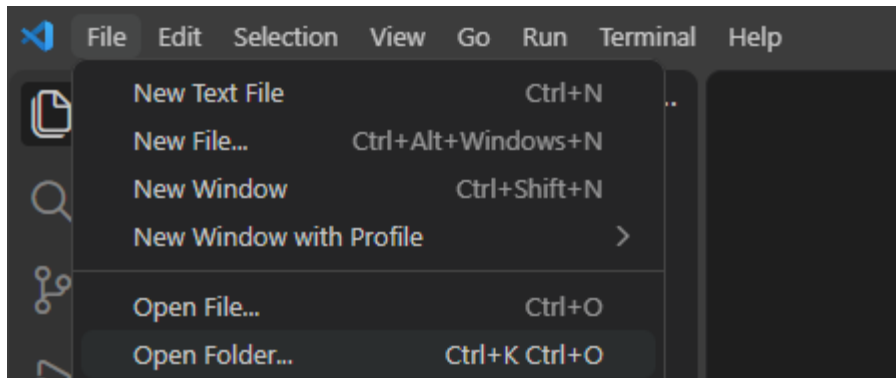
Once the connection is established, the Ewok host name should appear in the bottom-left corner of the VS Code window.



## 3. Binder Remodelling with Boltz

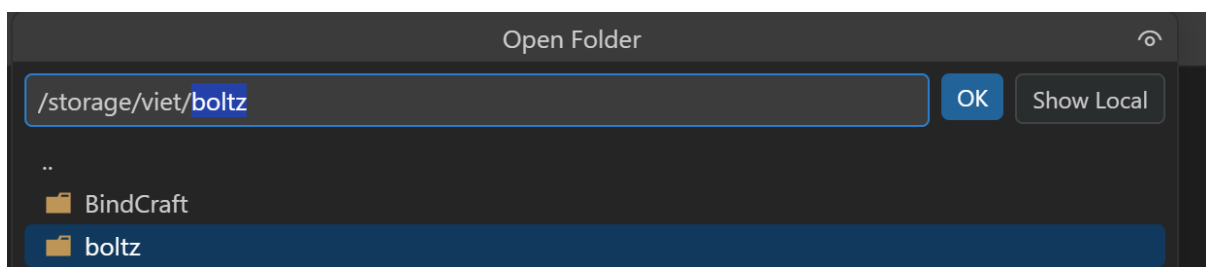
### 3.1 Open the Boltz Project

After connecting to the Ewok machine, select **File** → **Open Folder** in VS Code.



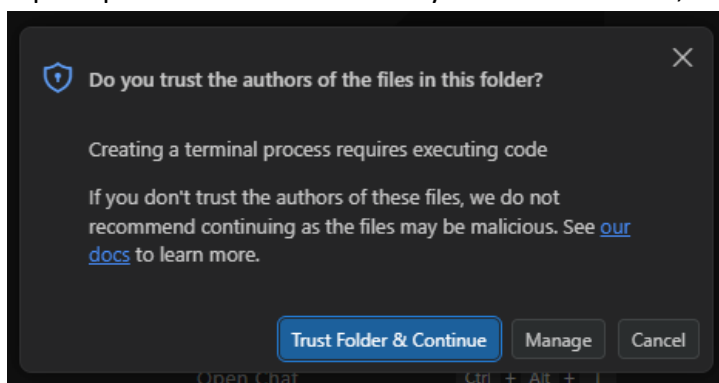
Open the directory in which Boltz is installed. Typical installation locations include:

- /home/<username>/Boltz
- /storage/<username>/Boltz



Replace <username> with your Ewok username.

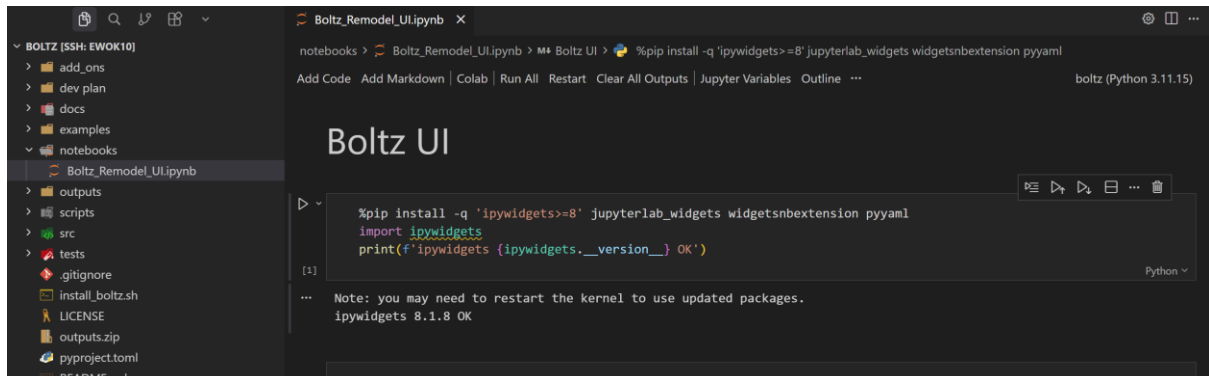
If prompted to confirm whether you trust the folder, select **Trust Folder & Continue**.



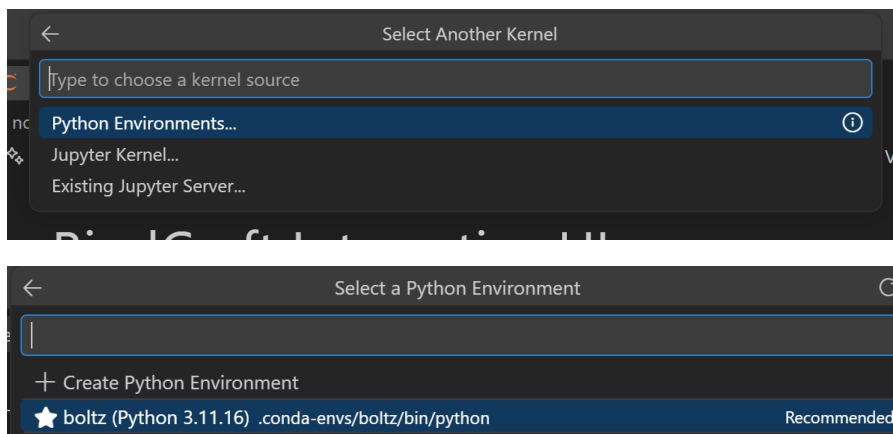
## 3.2 Boltz Remodel UI

### 3.2.1 Set Up the Notebook

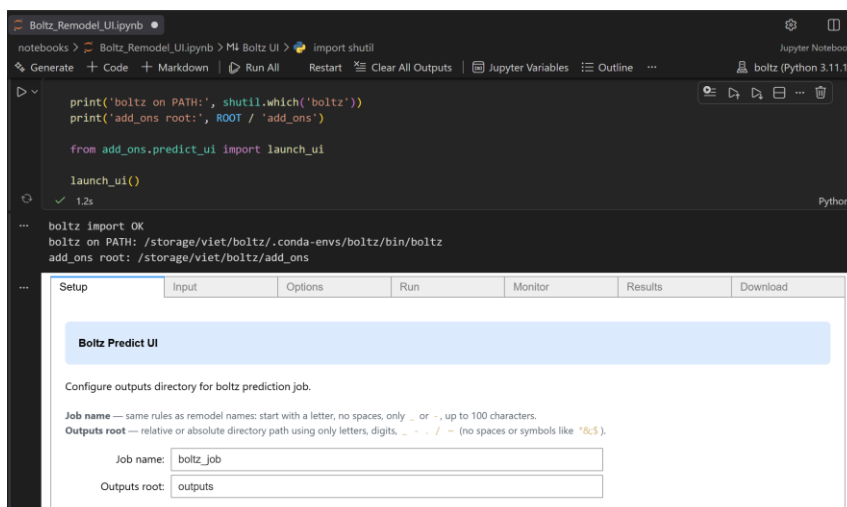
In the **Explorer** panel on the left side of VS Code, open notebooks/Boltz\_Remodel\_UI.ipynb



Click **Select Kernel**, choose **Python Environments**, and select the **Boltz** environment.



Then click **Run All** for dependencies installation and UI rendering.



If the UI does not appear, you may need to reload the notebook to ensure that the ipywidgets dependency is loaded correctly.

## 3.2.2 Run Binder Remodelling

### Step 1: Configure The Remodel Job

In the **Input** tab, configure your Binder remodel job. You can either load an existing configuration or create a new one.

To load an existing configuration, select it from the **Saved config** dropdown. To create a new configuration, leave the dropdown set to **select saved config**.

Input:

Binder remodel

**Edit or create a binder remodel config**

Pair one target with many binders — each binder becomes its own Boltz job.

Protein sequences supported only (no DNA, RNA, or ligands).

Saved config: IFIT5\_FL\_bindcraft\_remodel.json

Refresh list

**Target**

Tips for names (job, target, binder):

- For target, use something like: IFIT5\_cropped or binder-1
- For binder(s), use exact name(s) from BindCraft output, e.g. IFIT5\_bindcraft\_ewok-Binder145\_s470456\_rpmnd
- Start with a letter, no spaces, only \_ or - as separator, up to 100 characters

Tips for sequences:

- Paste a protein sequence (standard amino acids: ACDEFGHIKLMNPQRSTVWY)
- Letters only — numbers, spaces, and symbols are removed automatically
- Uppercased automatically: non-AA letters (e.g. S, J, O, U, X, Z) are highlighted; up to 500 residues

Job name: binder\_remodel\_example.json

A remodel job consists of one target and one or more binders. Each binder is processed as a separate Boltz job.

### Target Definition

| Field                  | Description                                                  |
|------------------------|--------------------------------------------------------------|
| <b>Job name</b>        | Name assigned to the Boltz job.                              |
| <b>Target name</b>     | Name of the target protein.                                  |
| <b>Target ID</b>       | Chain ID for the target in the predicted complex, usually A. |
| <b>Binder ID</b>       | Chain ID for the binder in the predicted complex, usually B. |
| <b>Target sequence</b> | Amino acid sequence of the target protein.                   |

Binder remodel

**Edit or create a binder remodel config**

Pair one target with many binders — each binder becomes its own Boltz job.

Protein sequences supported only (no DNA, RNA, or ligands).

Saved config: — select saved config —

Refresh list

**Target**

Tips for names (job, target, binder):

- For target, use something like: IFIT5\_cropped or binder-1
- For binder(s), use exact name(s) from BindCraft output, e.g. IFIT5\_bindcraft\_ewok-Binder145\_s470456\_rpmnd
- Start with a letter, no spaces, only \_ or - as separator, up to 100 characters

Tips for sequences:

- Paste a protein sequence (standard amino acids: ACDEFGHIKLMNPQRSTVWY)
- Letters only — numbers, spaces, and symbols are removed automatically
- Uppercased automatically: non-AA letters (e.g. S, J, O, U, X, Z) are highlighted; up to 500 residues

Job name: e.g. IFIT5\_cropped\_remodel

Target name: e.g. IFIT5\_cropped

Target id: e.g. A

Binder id: e.g. B

Target seq: Paste target protein sequence (letters A-Z only)..

## Binder Definition

Each binder requires the following fields:

| Field              | Description                        |
|--------------------|------------------------------------|
| <b>Binder name</b> | Name assigned to the binder.       |
| <b>Sequence</b>    | Amino acid sequence of the binder. |

## Manage Binders

Enter the number of binders and click **Update binder list**. Use **Prev** and **Next** to navigate pages, with up to five binders per page.

### Binders

Same naming and sequence constraints as above for every binder.

Set how many binders you need, then click **Update binder lists**. If the list is long, use Prev/Next (shows 5 at a time) or scroll.

Check binders individually, or use **Select page** for everyone on this page (selection is kept across pages). **Clear selection** unchecks all. **Delete selected** removes checked binders from the list.

When this page has fewer than 5 binders, use **Add binder** to append one without changing # binders.

# binders:

← Prev

Page 1 / 1 (1 binders, 5/page)

→ Next

☒ Select page

☐ #1 of 1

Binder 1:

Sequence:

Use **Add binder** to add one (max 5 per page). To remove binders, select each via checkbox or select all on a page via **Select page**, then click **Delete selected**. Use **Clear selection** to unselect all.

## Binders

Same naming and sequence constraints as above for every binder.

Set how many binders you need, then click **Update binder lists**. If the list is long, use Prev/Next (shows 5 at a time) or scroll.

Check binders individually, or use **Select page** for everyone on this page (selection is kept across pages). **Clear selection** unchecks all. **Delete selected** removes checked binders from the list. When this page has fewer than 5 binders, use **Add binder** to append one without changing # binders.

# binders:  Update binder list

← Prev
Page 1 / 3 (11 binders, 5/page, 5 selected)
→ Next

Select page
Clear selection (5)
Delete selected (5)

☒ #1 of 11

Binder 1:

Sequence:

☒ #2 of 11

Binder 2:

Sequence:

☒ #3 of 11

Binder 3:

Sequence:

## Save Config

Finally, to apply edit to the loaded config or create new config, enter a file name and click **Save config**. Configurations are saved as JSON files in `add_ons/configs/`

### Save

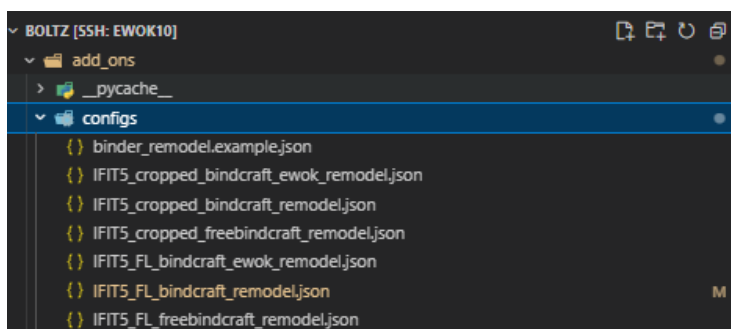
Pick a file name (same naming rules applied). Config files always saved under `add_ons/configs`.

File name:

Save config

Validation OK — saved `add_ons/configs/IFIT5_FL_bindcraft_remodel.json`

binder\_remodel target=IFIT5\_FL (len=482) binders=1 job\_name=IFIT5\_FL\_bindcraft\_remodel — 1 YAML file(s).



## Step 2: Choose Prediction Options

In the **Options** tab, choose how Boltz should run. If you are unsure, keep the default settings.

For full details, see <https://github.com/jwohlwend/boltz/blob/main/docs/prediction.md#options>.

**Prediction options**

Model: Boltz-2

Device: GPU 1: NVIDIA GeForce RTX 4090 — 22909 MiB free / 24564 MiB

[Refresh GPU list](#)

Sets `CUDA_VISIBLE_DEVICES` for the Boltz job. Memory values are point-in-time snapshots; use Refresh GPU list to update them.

Samples: 5

Recycling: 10

Sampling: 200

Format: mmCIF

Seed (-1=none): -1

☒ Use MSA server (`--use_msa_ser...`)

☒ Use potentials (`--use_potentials`)

☒ Override existing (`--override`)

☒ Write full PAE (`--write_full_pae`)

| Option            | Description                                                                             |
|-------------------|-----------------------------------------------------------------------------------------|
| <b>Model</b>      | Choose <b>Boltz-1</b> or <b>Boltz-2</b> .                                               |
| <b>Device</b>     | Select the GPU to use. Click <b>Refresh GPU list</b> to update latest memory available. |
| <b>Samples</b>    | Number of structure predictions generated for each binder.                              |
| <b>Recycling</b>  | Number of refinement rounds.                                                            |
| <b>Sampling</b>   | Number of calculation steps for each structure.                                         |
| <b>Format</b>     | Choose mmCIF or PDB for the output structure.                                           |
| <b>Seed</b>       | Leave as -1 unless you need repeatable results.                                         |
| <b>MSA server</b> | Automatically prepares the sequence information Boltz needs.                            |
| <b>Potentials</b> | Adds extra guidance during prediction.                                                  |
| <b>Override</b>   | Override existing outputs with same job name.                                           |
| <b>PAE</b>        | Saves an additional predicted alignment error file.                                     |



### Step 5: Results Inspection

After the job finishes, results are saved in the outputs/ folder. Each binder has its own results folder: outputs/<job\_name>/out/boltz\_results\_yaml/predictions/<binder\_name>/

For N samples (set in **Options** tab), Boltz creates N predictions: model\_0 to model\_N-1. model\_0 is the highest-confidence prediction.

Each sample prediction will have three main files:

| File              | Description                                        |
|-------------------|----------------------------------------------------|
| .cif or .pdb      | Predicted 3D structure.                            |
| confidence_*.json | Confidence scores shown in the <b>Results</b> tab. |
| pae_*.npz         | Pairwise confidence/error data.                    |

Open the **Results** tab and select a job to view its predictions. Results are sorted by **ipTM**.

See the official <https://github.com/jwohlwend/boltz/blob/main/docs/prediction.md#output>.

Setup
Input
Options
Run
Monitor
**Results**
Download

### Results

Pick a job folder under **outputs/** to inspect prediction metrics (sorted by ipTM).

Output folder: IFIT5\_cropped\_bindcraft\_ewok\_remodel

Loaded 55 result row(s) from /storage/viet/boltz/outputs/IFIT5\_cropped\_bindcraft\_ewok\_remodel/out/boltz\_results\_yaml/predictions

Results (sorted by ipTM)

| design                                        | model | ipTM   | confidence_score | ptm    | complex_pliddt | protein_ipTM | ligand_ipTM |
|-----------------------------------------------|-------|--------|------------------|--------|----------------|--------------|-------------|
| IFIT5_bindcraft_ewok-Binder_193_s794158_mpnn1 | 3     | 0.956  | 0.9332           | 0.9552 | 0.9275         | 0.956        | 0.0         |
| IFIT5_bindcraft_ewok-Binder_193_s794158_mpnn1 | 2     | 0.9518 | 0.935            | 0.9547 | 0.9308         | 0.9518       | 0.0         |
| IFIT5_bindcraft_ewok-Binder_193_s794158_mpnn1 | 0     | 0.951  | 0.9373           | 0.9537 | 0.9339         | 0.951        | 0.0         |
| IFIT5_bindcraft_ewok-Binder_193_s794158_mpnn3 | 3     | 0.9475 | 0.9133           | 0.9514 | 0.9048         | 0.9475       | 0.0         |

```

outputs
├── IFIT5_cropped_bindcraft_ewok_remodel
│   ├── out / boltz_results_yaml
│   │   ├── lightning_logs
│   │   ├── msa
│   │   └── predictions
│   │       ├── IFIT5_bindcraft_ewok-Binder_172_s275174_mpnn13
│   │       │   ├── confidence_IFIT5_bindcraft_ewok-Binder_172_s275174_mpnn13_model_
│   │       │   │   ├── confidence_IFIT5_bindcraft_ewok-Binder_172_s275174_mpnn13_model_
│   │       │   │   ├── confidence_IFIT5_bindcraft_ewok-Binder_172_s275174_mpnn13_model_
│   │       │   │   ├── confidence_IFIT5_bindcraft_ewok-Binder_172_s275174_mpnn13_model_
│   │       │   │   ├── confidence_IFIT5_bindcraft_ewok-Binder_172_s275174_mpnn13_model_
│   │       │   │   ├── IFIT5_bindcraft_ewok-Binder_172_s275174_mpnn13_model_0.cif
│   │       │   │   ├── IFIT5_bindcraft_ewok-Binder_172_s275174_mpnn13_model_1.cif
│   │       │   │   ├── IFIT5_bindcraft_ewok-Binder_172_s275174_mpnn13_model_2.cif
│   │       │   │   ├── IFIT5_bindcraft_ewok-Binder_172_s275174_mpnn13_model_3.cif
│   │       │   │   ├── IFIT5_bindcraft_ewok-Binder_172_s275174_mpnn13_model_4.cif
│   │       │   │   ├── pae_IFIT5_bindcraft_ewok-Binder_172_s275174_mpnn13_model_0.npz
│   │       │   │   ├── pae_IFIT5_bindcraft_ewok-Binder_172_s275174_mpnn13_model_1.npz
│   │       │   │   ├── pae_IFIT5_bindcraft_ewok-Binder_172_s275174_mpnn13_model_2.npz
│   │       │   │   ├── pae_IFIT5_bindcraft_ewok-Binder_172_s275174_mpnn13_model_3.npz
│   │       │   │   ├── pae_IFIT5_bindcraft_ewok-Binder_172_s275174_mpnn13_model_4.npz
│   │       │   │   ├── pde_IFIT5_bindcraft_ewok-Binder_172_s275174_mpnn13_model_0.npz
│   │       │   │   ├── pde_IFIT5_bindcraft_ewok-Binder_172_s275174_mpnn13_model_1.npz
│   │       │   │   ├── pde_IFIT5_bindcraft_ewok-Binder_172_s275174_mpnn13_model_2.npz
│   │       │   │   ├── pde_IFIT5_bindcraft_ewok-Binder_172_s275174_mpnn13_model_3.npz
│   │       │   │   ├── pde_IFIT5_bindcraft_ewok-Binder_172_s275174_mpnn13_model_4.npz
│   │       │   │   ├── pliddt_IFIT5_bindcraft_ewok-Binder_172_s275174_mpnn13_model_0.npz
│   │       │   │   ├── pliddt_IFIT5_bindcraft_ewok-Binder_172_s275174_mpnn13_model_1.npz
│   │       │   │   ├── pliddt_IFIT5_bindcraft_ewok-Binder_172_s275174_mpnn13_model_2.npz
│   │       │   │   ├── pliddt_IFIT5_bindcraft_ewok-Binder_172_s275174_mpnn13_model_3.npz
│   │       │   │   ├── pliddt_IFIT5_bindcraft_ewok-Binder_172_s275174_mpnn13_model_4.npz
│   │       │   ├── IFIT5_bindcraft_ewok-Binder_172_s275174_mpnn18
│   │       │   └── IFIT5_bindcraft_ewok-Binder_176_s389396_mpnn1

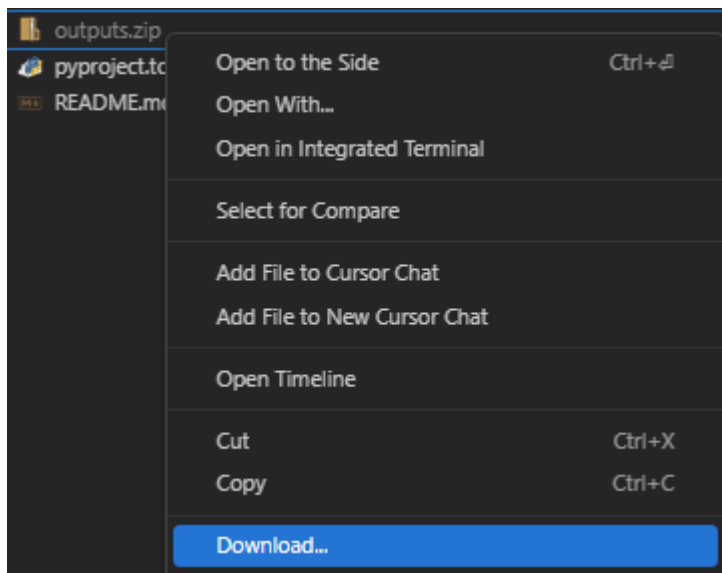
```

### Step 6: Download Boltz outputs

Use the **Download** tab to create a ZIP file, especially when downloading many binders or large sequences.

| Setup                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Input | Options | Run | Monitor | Results | Download |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|---------|-----|---------|---------|----------|
| <div>Zip outputs for download</div> <p>Choose the entire <b>outputs/</b> folder or a single design folder. The archive is written to the project root (parent of <code>notebooks/</code>): <code>/storage/viet/boltz/</code>.</p> <p>Zip source: <input type="text" value="Entire outputs/ folder"/></p> <p>Zip file: <input type="text" value="outputs.zip"/></p> <p><a href="#">Refresh folders</a></p> <p><a href="#">Create zip in project root</a></p> <p>Created <code>/storage/viet/boltz/outputs.zip</code> (569.89 MB)</p> |       |         |     |         |         |          |

In the Explorer panel, at project root **right click** on the ZIP Boltz output then **Download**.

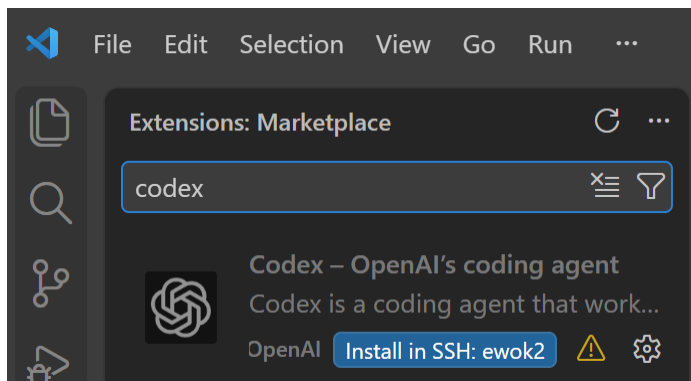


## 4. Useful Extensions

### Codex

If you have a ChatGPT account, you can optionally install the Codex extension on the Ewok host for AI-assisted work within the project.

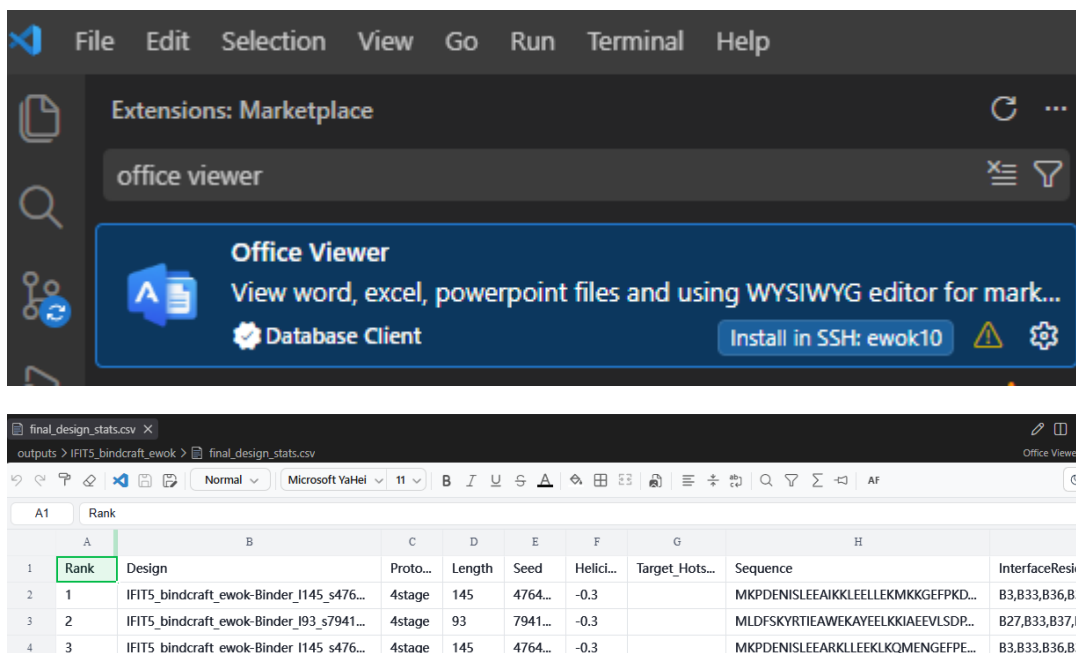
Extension: <https://marketplace.cursorapi.com/items/?itemName=openai.chatgpt>



### Office Viewer

This extension helps with reading BindCraft output csv files directly in VS Code.

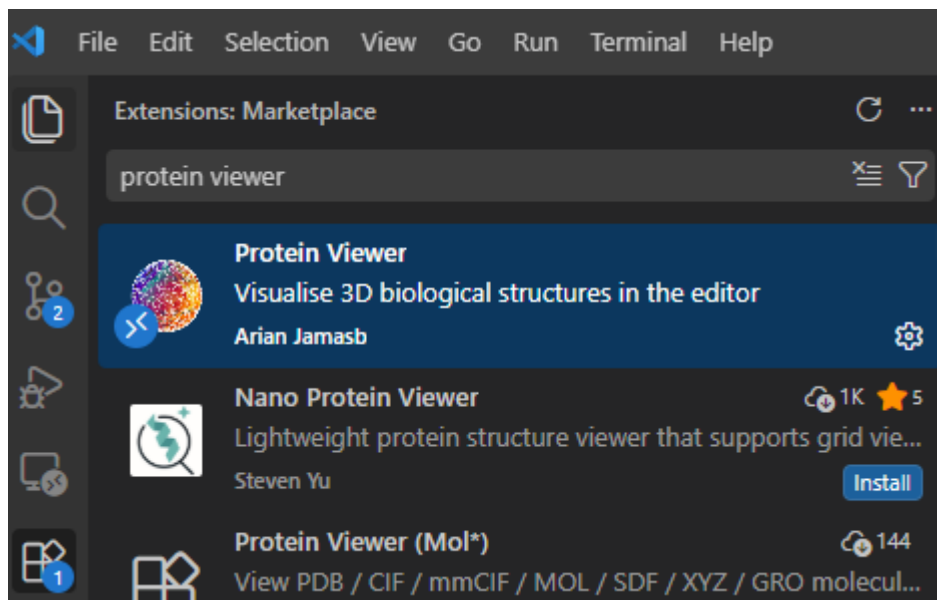
Extension link: <https://marketplace.cursorapi.com/items/?itemName=cweijan.vscode-office>



## Protein Viewer

This allows inspection of target protein or generated binders PDB directly in VS Code.

Extension link: <https://marketplace.visualstudio.com/items?itemName=ArianJamasp.protein-viewer>



Right click on a PDB and select *Launch Protein Viewer* to view its 3D structure.

